

Section 2: Byte-Pair Tokenizer

2.1 The Unicode Standard

(a)

```
>>> chr(0)
'\x00'
```

(b)

The printed representation returns an empty line, while the character's string representation returns

```
>>> chr(0)
'\x00'
```

(c)

`chr(0)` encodes a NULL byte, so nothing is printed in its position when it occurs in printed text.

2.2 Unicode Encodings

(a)

UTF-8 encoded bytes are shorter per character; UTF-32 can have as many as 8 bytes per Unicode character. The byte vocabulary in both UTF-32 and UTF-16 is also much larger than the manageable 256.

(b)

Passing

```
>>> aigue_encoded = "é".encode("utf-8")
```

to the incorrect function yields a `UnicodeDecode` error. This decode function assumes that one byte corresponds to a Unicode character, which is not always true.

(c)

The bytestring `b'\xe3\x81'` doesn't correspond to any Unicode character; it is a subset of `b'\xe3\x81\x93'` which corresponds to 'ㄣ'